



CARBON FOOTPRINT CALCULATOR



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TABLE OF CONTENTS

1. WHAT IS CARBON FOOTPRINT?

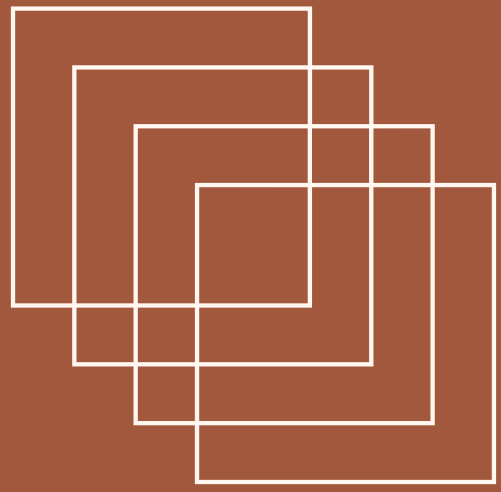
2. WHAT IS THE PROBLEM? WHY DO WE NEED A CALCULATOR?

3. OUR WEBSITE

4. SUSTAINABILITY ANALYSIS

5. HOW CAN WE REDUCE EMISSIONS?

6. CONCLUSION & FUTURE IMPROVEMENTS



ENVIRONMENTAL PROBLEM: HIGH CARBON EMISSIONS

- Increasing greenhouse gas emissions
- Major contributors: transportation, electricity, industries, food, and waste
- Leads to global warming and climate change
- Need for awareness and reduction of individual emissions



WHAT IS CARBON FOOTPRINT?

The total amount of greenhouse gases released into the atmosphere as a result of our activities, usually expressed as carbon dioxide equivalent (CO₂e).

- Mainly associated with carbon dioxide (CO₂) and other greenhouse gases
- Measured in CO₂ equivalent (CO₂e)
- Generated directly or indirectly by our daily activities

Examples:

- Transportation
- Electricity usage
- Food consumption
- Production and purchase of goods
- Waste generation

THE PROBLEM: HIGH CARBON EMISSIONS

OUR EVERYDAY CHOICES LEAVE A CARBON FOOTPRINT

DAILY ACTIVITY

TRANSPORTATION



ELECTRICITY AND
APPLIANCES



FOOD PRODUCTION AND
WASTE



LANDFILLS



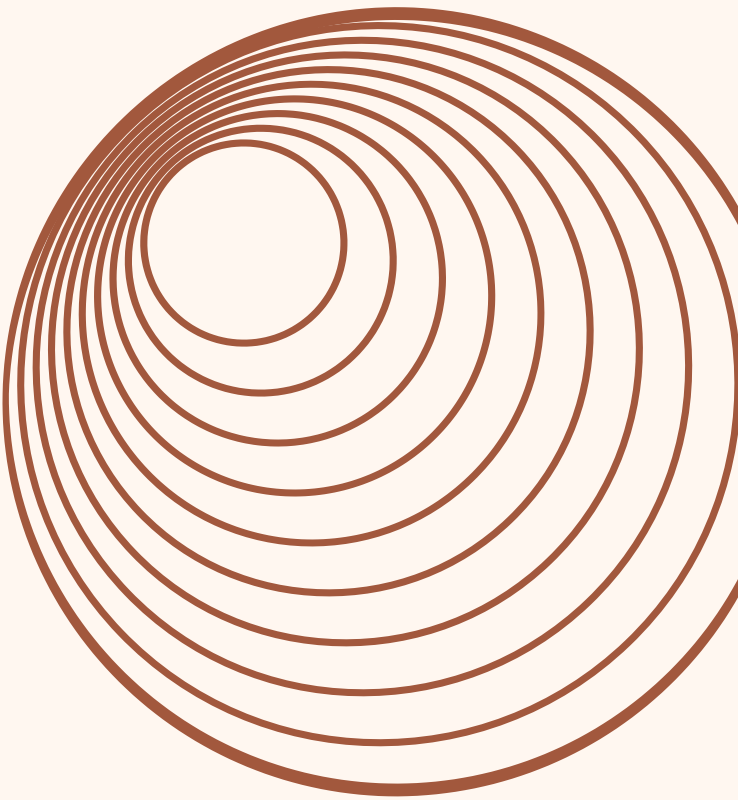
CO2 EMISSIONS

DIRECT CO2 AND OTHER GREEN HOUSE GASES
RELEASE

INDIRECT EMISSIONS THROUGH HEATING & COOLING

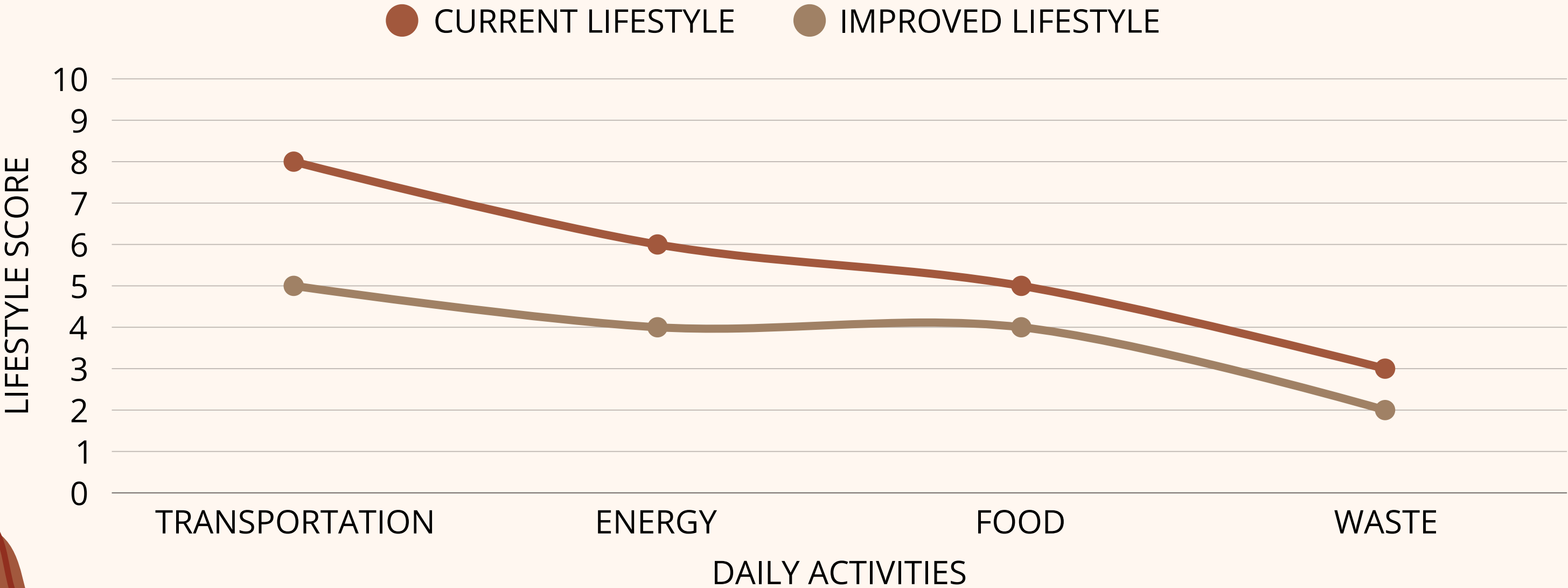
DIRECT GREENHOUSE GAS EMISSIONS

PRODUCES CO2 & METHANE AS IT
DECOMPOSES



WHY DO WE NEED A CARBON FOOTPRINT CALCULATOR?

- A CALCULATOR CONVERTS EVERYDAY ACTIVITIES INTO A MEASURABLE CARBON/LIFESTYLE SCORE
- THE GRAPH SHOWS WHICH LIFESTYLE AREAS CONTRIBUTE THE MOST. A HIGHER SCORE MEANS THAT ACTIVITY HAS A GREATER RELATIVE IMPACT, MAKING THE BIGGEST CONTRIBUTORS EASY TO SPOT.
- ONCE HIGH-IMPACT ACTIVITIES ARE IDENTIFIED, USERS CAN COMPARE THEIR CHOICES AND FOCUS ON THE AREAS WHERE REDUCING EMISSIONS WILL HAVE THE GREATEST EFFECT.



CarbonTrack — Our Website

- An interactive carbon-footprint calculator for college students
- The website asks users about their daily habits in five categories:
 - Food & Drink: Dietary and food-related habits.
 - Transportation: Travel modes and distance.
 - Electricity: Electricity usage and consumption habits.
 - Waste: Waste-generation and recycling habits.
 - Everyday Consumption: Clothing, electronics and online purchases.
- **Technology Used**
 - HTML & CSS: Website structure, design and mobile compatibility.
 - JavaScript: Processes inputs, performs calculations and handles interactions

CarbonTrack — Our Website

- **How It Works & What It Shows?**

User inputs → JavaScript processing → Emission estimation → Results dashboard

The user completes the questionnaire using sliders and selectable options.

JavaScript applies simplified emission estimates to the collected information.

The website calculates an approximate monthly carbon footprint in kg CO₂e.

It compares the emissions from each category to identify the largest contributor.

The website is an educational prototype using approximate estimates, not a scientifically exact carbon-accounting system.

CarbonTrack — Our Website

- **What does the results page display?**

Total estimated monthly footprint

Category-wise emission breakdown

Largest emission contributor

Personalized recommendations

Dynamic recommendations: displays suggestions related to users largest emission category

WAYS TO REDUCE CARBON EMISSION

- **Use public transportation**

Instead of using private vehicles for every journey, we can use buses, trains, carpooling, cycling, or walking whenever possible.

- **Choose Cleaner Vehicles**

We can choose vehicles that produce fewer emissions. Correct tyre pressure, regular servicing and avoiding unnecessary idling can improve fuel efficiency.

- **Reduce Electricity Consumption**

Air conditioners can also be used responsibly by setting a suitable temperature and keeping doors and windows closed while they are running.

- **Use Renewable Energy**

Solar and wind energy produce electricity with much lower carbon emissions than fossil-fuel-based energy.

WAYS TO REDUCE CARBON EMISSION

- **Reduce, Reuse and Recycle**

We should reduce unnecessary consumption. We should reuse products whenever possible instead of throwing them away after one use. Materials such as paper, plastic, glass and metal should be separated and recycled properly.

- **Make Sustainable Shopping Choices**

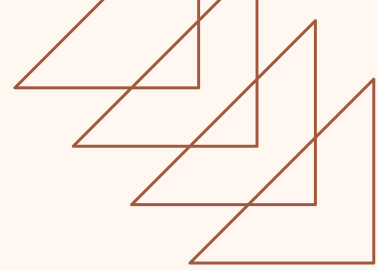
We can avoid buying products that we do not really need and choose durable products that can be used for a longer period. Carrying reusable shopping bags instead of using disposable ones is another simple step.

- **Save Water**

Water needs to be pumped, treated, heated and transported, and all these processes require energy. We can close taps when they are not needed, repair leaking taps and pipes, take shorter showers and avoid unnecessary water usage.

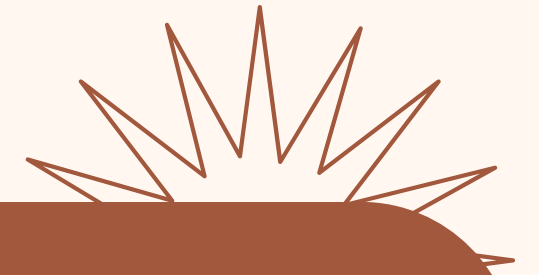
- **Use Digital Resources Wisely**

We can avoid keeping unnecessary devices switched on and unplug chargers when they are not being used. We can also extend the life of electronic devices by taking proper care of them instead of replacing them unnecessarily.



FUTURE IMPROVEMENTS

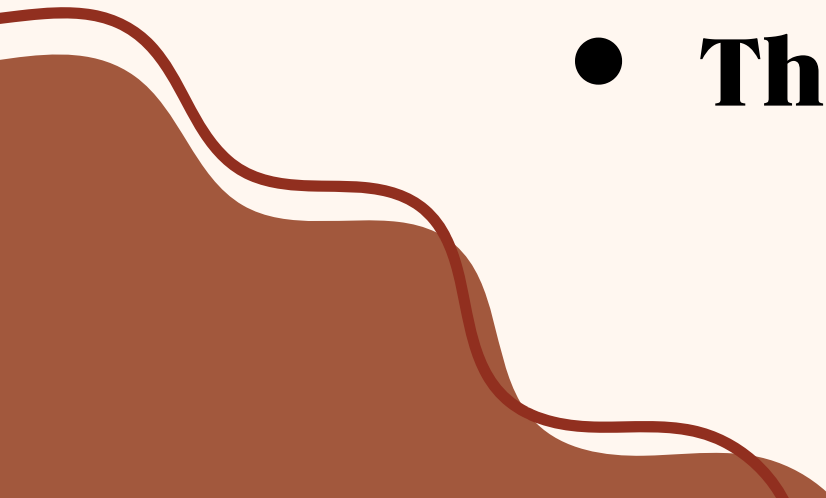
Towards a Lower-Carbon Future



- Greater use of renewable energy such as solar and wind.
- Expansion of public transportation and electric mobility.
- Development of energy-efficient technologies and buildings.
- Better waste management, recycling and circular use of resources.
- More sustainable practices in industries and agriculture.
- Increased awareness and participation of individuals and communities..



CONCLUSION

- **CarbonTrack helps users understand their personal carbon footprint.**
 - **It converts everyday lifestyle choices into an estimated carbon emission value.**
 - **The category-wise results help identify the major sources of emissions.**
 - **Personalized recommendations encourage users to make more sustainable choices.**
 - **The main goal is to turn awareness into action**
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THANK
YOU

